

BALLU HARISH
Web - GIS / WEB Developer
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Carrer Objective

Seeking an opportunity within a dynamic and challenging environment where I can leverage my extensive experience and skills in web development and GIS technologies to make meaningful contributions towards the success of the organization. With a strong foundation in both front-end and back-end web development, coupled with expertise in GIS tools and techniques, I aim to not only drive innovative solutions but also continuously enhance my personal skills. I am eager to collaborate with a team of talented professionals to create cutting-edge web applications and spatially enabled solutions that address complex challenges and propel the organization forward in its mission. Total industrial experience is 6+ years in Web GIS/ Web development environment .

Organizational Experience

Project Scientist

INCOIS, Pragathi Nagar, Hyderabad

September 2024 – Present

I am currently working as a Project Scientist at INCOIS, focusing on web application development and system modernization. I am involved in migrating an existing web platform from a legacy environment to a modern, efficient infrastructure. The earlier system used **WebSphere** as the web server, **Oracle** as the database, and **ArcGIS Server** for geospatial data management. My role involves transitioning the system to **Apache Tomcat and Apache HTTP Server, MySQL** for database management, and **GeoServer** for geospatial services. During this migration, I addressed challenges related to deprecated technologies through careful planning and execution, ensuring seamless functionality with minimal disruption. I have also focused on performance optimization, system compatibility, and overall stability of the migrated platform.

Research Scientist

NRSC-ISRO, Bala Nagar, Hyderabad

September 2022 – March 2024

As a Research Scientist at NRSC-ISRO, I led the **front-end development of the Bhuvan Panchayat Portal**, focusing on delivering a responsive and user-friendly interface. I utilized **React JS, Bootstrap, jQuery, and AJAX** to enhance user experience and ensure cross-device compatibility. I also managed server-side components including **GeoServer, PostgreSQL/PostGIS, and Node.js**, ensuring smooth data delivery. My responsibilities included developing and maintaining APIs, including **GraphQL queries**, implementing **Python and Shell scripts** for data processing and server automation, and integrating large geospatial datasets. Key challenges included automating backend services and making the portal fully responsive, both of which were successfully addressed.

GIS Developer

JNTUH, Hyderabad

February 2016 – September 2020

During my tenure as a GIS Developer, I spearheaded the project titled **“Exchange of Geospatial Data through Geo-web”**, which focused on accessing, visualizing, and publishing geospatial data through web platforms. I worked extensively with **ESRI ArcGIS Server** services, including Map, Feature, and Image services, and designed interactive web pages using **stylesheets and the Dojo Toolkit**. I developed a comprehensive map-view environment with custom rendering tools, pop-up templates, query tasks, widgets, and layer toggles to enhance data visualization and user interaction. This work significantly improved the efficiency and usability of geospatial data exchange.

Associate Photogrammetry

Avineon India Pvt. Ltd., Hyderabad

March 2015 – January 2016

At Avineon India Pvt. Ltd., I contributed to multiple international geospatial projects. For the **Saudi Telecom Company**, I used **ArcGIS and ERDAS Imagine** to identify regions lacking telecom tower coverage, supporting strategic infrastructure planning. For **IGN-France**, I worked on a pilot project using **MicroStation, TerraScan, and TerraModeler** to compute bare-earth surfaces for environmental assessment and planning. These projects strengthened my expertise in photogrammetry, satellite imagery processing, and geospatial analysis.

Certifications

- Certified in Microsoft Digital Literacy Test on March-03-2011.
- Certified in IIRS 18th Outreach program “Basics of Remote Sensing, Geographical Information System & Global Navigation Satellite System” conducted by CIST-IST-JNTUH during August-November 2016.
- Certified in IIRS 19th Outreach program “Remote Sensing and GIS Applications in Carbon Forestry” conducted by CIST-IST-JNTUH during February 16, 2017 to March 10, 2017.
- Certified in IIRS 20th Outreach program “Microwave Radar Remote Sensing and its Applications” conducted by CIST-IST-JNTUH from April 10, 2017 to April 27, 2017.

- Certified in IIRS 21st Outreach program” “RS & GIS Applications in Water Resource Management” conducted by CIST-IST-JNTUH from May 22, 2017 to June 09, 2017.
- Certified in IIRS 25th Outreach program” “Hyperspectral Remote Sensing and its Applications” conducted by CIST-IST-JNTUH from September 18, 2017 to September 21, 2017”
- Certified in IIRS 33rd Outreach programme “Advance Image Analysis” conducted by CIST-IST-JNTUH from June-04-2018 to June-15- 2018”
- Certified in IIRS Outreach programme “Geospatial Inputs for Enabling Master Plan Formulation under AMRUT Sub-scheme”” conducted by CIST-IST-JNTUH from July 23, 2018 to July 27, 2018.
- Certified in IIRS Outreach programme “Geoprocessing using python” conducted by CIST-IST-JNTUH from July 22 2019 to 26 July 2019
- Certified in IIRS Outreach programme “Web GIS- Geo-visualization and online mapping” conducted by CIST-IST-JNTUH from 25-11-2019 to 29-11-2019.
- Certified in IIRS Outreach programme “Overview of Planetary Geosciences with special emphasis to the Moon and Mars”.conducted by CIST-IST-JNTUH from 08-06-2020 to 12-06-2020.
- Certified in IIRS Outreach programme “Geoprocessing using python” conducted by CIST-IST-JNTUH from July 22 2019 to 26 July 2019

Remarkable Achievements

- Authored a book chapter titled “Internet Technologies” (Second Chapter) in the edited volume Advances in Geospatial Technologies for Natural Resource Management, published by CRC Press, Taylor & Francis Group, 2024.
- Appointed as a Editor for the book titled “Futuristic Tends in Information Technology” published by IIP, Book Series ID: IIP-V2-2022-BS-14-02
- Published a chapter titled “Displaying the Map server data using Arc-GIS server Javascript API” in the book Research Development in Science & Technology

Qualification

- M.Tech(Spatial IT) from JNTUH-IST with 70.38% in the year 2014.
- B.Tech. (CSE) from Aizza college of Engg& Technology (JNTUH) With 67.73%, in the year 2011.
- Intermediate (M.P.C) from Board of Intermediate Education (AndhraPradesh) with 90% in the year 2007.
- S.S.C from Board of Secondary Education (Andhra Pradesh), with 76.3%, in the year2005.

Academic Projects

Project:

Company Name: Indian National Centre for Ocean Information Services.

Project Name : Automation of GTS data processing

Programming

Language's : Java

Database : SQL server

Software's : Chart director, UMN map server, open layers.

Hardware : Intel Pentium O/s: Windows 2007.

Client : INCOIS

Work role : Project student (M.Tech).

Description: This project involves developing of intranet application for INCOIS which includes sharing of weather information to the end users in pictorial format and also data can be downloaded in spread sheet format for certain time interval specified by user.

My Role: My role in this project is downloading of GTS data from RTH hub (New Delhi) after the downloading it need to be decrypted which is done using JAVA. After the decryption the data need to pushed to database (SQL) which is done using Java. Whenever user requests weather information regarding particular time interval he will have two options i.e viewing of information in pictorial format this is done using chart director and the next is downloading of data in spread sheet format. I worked for a period of 1 year.

Technical Skills

- Programming Languages : C, C++, Java, C#.NET, Python
- Scripting Languages : HTML, CSS, JavaScript, Bootstrap
- UI/UX & Frontend Technologies : UI/UX Design, jQuery, AJAX, Chart Director
- Web Technology Stacks : MERN Stack, MEAN Stack
- Databases : Oracle, MS SQL Server, PostgreSQL, PostGIS.
- GIS Servers & APIs : ArcGIS Server, ArcGIS Server JavaScript API, GeoServer, MapServer
- Web GIS Libraries : OpenLayers, Leaflet
- GIS & Spatial Tools : ArcPy, ArcObjects, ArcGIS, QGIS
- Image Processing Software : ERDAS Imagine

Personal Details

Date of Birth: 12th August 1990
Languages Known: English, Telugu and Hindi
Address: PragathiNagar Kukatpally Hyderabad.